10.13 Worked Example: Tree-Level φ^3 Scattering and a Two-Body Decay

LecNote10 derives $d\sigma$ and $d\Gamma$ (pp. 5–10) and ends with $d\sigma/d\Omega$ and $d\sigma/dt$ (p. 11), but never puts an amplitude in. So a student asks: what does an actual cross section look like? Does the $\frac{1}{2}$ for identical particles go into $d\sigma/d\Omega$, or only into σ? How big is it in barns, and how does it behave at threshold and at high energy? What is the decay rate of an actual particle? For $\varphi\varphi \to \varphi\varphi$ at tree level the answer is the closed form (10.13.5); a two-body decay follows.

Conventions are Srednicki's: $g_{\mu\nu} = \mathrm{diag}(-,+,+,+)$, $k^2 = -m^2$ on shell, $\tilde{\Delta}(k) = 1/(k^2 + m^2 - i\epsilon)$, $\langle 0|T\varphi(x)\varphi(y)|0\rangle = \frac{1}{i}\Delta(x-y)$, $\mathcal{L}_1 = \frac{1}{6}Z_g g\varphi^3 + \mathcal{L}_{\rm ct}$ in $d = 4$, so g is a mass; at tree level $Z_g = 1$ and $\mathcal{L}_{\rm ct}$ does not enter. $s = -(k_1+k_2)^2$, $t = -(k_1-k_1')^2$, $u = -(k_1-k_2')^2$, $s+t+u = 4m^2$. $\mathcal{T}$ is the amplitude, T a time.

From $\langle f|i\rangle$ to σ, in brief

For $f \neq i$, $\langle f|i\rangle = (2\pi)^4\delta^4(k_{\rm in} - k_{\rm out})\, i\mathcal{T}$ (LecNote10 p. 5). Its square carries $(2\pi)^4\delta^4(0) = \int d^4x = VT$ for a box of volume V watched for a time T (p. 6). Plane waves have $\langle k|k\rangle = (2\pi)^3 2\omega\, \delta^3(\mathbf{0}) = 2\omega V$, hence the norms in $P = |\langle f|i\rangle|^2/(\langle f|f\rangle\langle i|i\rangle)$. Each final particle brings $V\, d^3k'/(2\pi)^3$ states and $1/(2\omega' V)$, together $d\tilde{k}'$; the initial pair brings $1/(4E_1E_2V^2)$. So $dP \propto VT/V^2 = T/V$ (p. 7): a constant rate, whose $1/V$ (one beam particle per box) the flux $|\mathbf{k}_1|/(E_1V)$ removes (p. 8). That leaves $4|\mathbf{k}_1|E_2$, which in the target's rest frame is $4|\mathbf{k}_1^{\rm FT}|m_2 = 4|\mathbf{k}_1^{\rm CM}|\sqrt{s}$, the flux factor (pp. 8–9). For a decay $\langle i|i\rangle = 2E_1V$ removes the V, and there is no flux (p. 10). With $d\mathrm{LIPS} = (2\pi)^4\delta^4(k_{\rm in} - k_{\rm out})\prod_j d\tilde{k}_j'$ (pp. 9–11),

$$d\sigma = \frac{|\mathcal{T}|^2\, d\mathrm{LIPS}_2}{4|\mathbf{k}_1^{\rm CM}|\sqrt{s}}, \qquad d\Gamma = \frac{|\mathcal{T}|^2\, d\mathrm{LIPS}}{2E_1}, \qquad d\mathrm{LIPS}_2\big|_{\rm CM} = \frac{|\mathbf{k}_1'|\, d\Omega}{16\pi^2\sqrt{s}}. \tag{10.13.1}$$

Why σ is the same in every frame is §10.12.

The amplitude and the physical region

The vertex is ig (the $3!$ ways to attach three lines cancel the $\frac{1}{6}$) and the propagator $\frac{1}{i}\tilde{\Delta}$. The three diagrams exchange momentum $q = k_1 + k_2$, $k_1 - k_1'$ and $k_1 - k_2'$, so $i\mathcal{T} = (ig)^2\,\frac{1}{i}\sum_q 1/(q^2 + m^2 - i\epsilon)$ (LecNote9 p. 8), and $(ig)^2/i = ig^2$. With $(k_1+k_2)^2 + m^2 = m^2 - s$, and likewise for t and u,

$$\mathcal{T} = g^2\Big[\frac{1}{m^2 - s} + \frac{1}{m^2 - t} + \frac{1}{m^2 - u}\Big] + O(g^4). \tag{10.13.2}$$

LecNote10 p. 4 has the opposite overall sign, which drops out of $|\mathcal{T}|^2$; the relative signs do not. In the CM frame $|\mathbf{k}|^2 = \frac{s}{4} - m^2$ for all four momenta, $t = -2|\mathbf{k}|^2(1-\cos\theta)$ and $u = -2|\mathbf{k}|^2(1+\cos\theta)$, so the physical region is $s \geq 4m^2$, $-(s-4m^2) \leq t, u \leq 0$. There $m^2 - s \leq -3m^2$ while $m^2 - t$, $m^2 - u \geq m^2$: the s-channel term is negative and the other two are positive, so the s-channel interferes destructively with the t- and u-channels. At threshold $\mathcal{T}_{\rm thr} = (g^2/m^2)(-\frac{1}{3}+1+1) = \frac{5}{3}g^2/m^2$. No denominator vanishes, so $\mathcal{T}$ is real and every $-i\epsilon$ can be set to zero. It would matter only if an exchanged particle could go on shell in the physical region; that case appears at the end.

Differential and total cross sections

Equal masses give $|\mathbf{k}_1'| = |\mathbf{k}_1|$ in (10.13.1). With $dt = 2|\mathbf{k}|^2 d\cos\theta$, and $d\Omega = 2\pi\, d\cos\theta$ once the azimuth is integrated (p. 11),

$$\frac{d\sigma}{d\Omega_{\rm CM}} = \frac{|\mathcal{T}|^2}{64\pi^2 s}, \qquad \frac{d\sigma}{dt} = \frac{|\mathcal{T}|^2}{16\pi s(s-4m^2)}, \tag{10.13.3}$$

both symmetric under $t \leftrightarrow u$, i.e. $\theta \to \pi - \theta$.

Where the $\frac{1}{2}$ goes. $|\mathbf{k}_1', \mathbf{k}_2'\rangle$ and $|\mathbf{k}_2', \mathbf{k}_1'\rangle$ are one state, so integrating over all directions of $\mathbf{k}_1'$ counts each final state twice, and $\sigma = \frac{1}{2}\int_{4\pi} d\Omega\,(d\sigma/d\Omega) = \int_{\cos\theta>0} d\Omega\,(d\sigma/d\Omega)$. This is p. 11's $S = 2$, and it goes into the total only. Eq. (10.13.3) carries no $\frac{1}{2}$, because it gives the rate at which a detector at θ counts particles: an event with one particle at Ω has the other at $-\Omega$, and both are counted. Nor is this S the symmetry factor of a diagram (§10.9), which is 1 for all three tree diagrams.

The integral. With $x = -t \in [0, L]$, $L \equiv s - 4m^2$, one has $m^2 - u = m^2 + L - x$, and the squares and cross terms of (10.13.2) need, besides $\int_0^L dx = L$, only ($\ell \equiv \ln[(s-3m^2)/m^2]$)

$$\int_0^L \frac{dx}{(m^2+x)^2} = \frac{L}{m^2(s-3m^2)}, \qquad \int_0^L \frac{dx}{m^2+x} = \ell,$$
$$\int_0^L \frac{dx}{(m^2+x)(m^2+L-x)} = \frac{2\ell}{s-2m^2}. \tag{10.13.4}$$

Then $\sigma = \frac{1}{2}\int dt\, d\sigma/dt$ is

$$\sigma = \frac{g^4}{32\pi s}\left[\frac{1}{(s-m^2)^2} + \frac{2}{m^2(s-3m^2)} + \frac{4m^2\,\ell}{(s-m^2)(s-2m^2)(s-4m^2)}\right]. \tag{10.13.5}$$

The three terms are the s-channel squared, the t- and u-channel squares, and the three cross terms (the negative s–t and s–u terms are outweighed by the positive t–u term). Direct quadrature of (10.13.3) reproduces it for s from $4.0001\,m^2$ to $10^6 m^2$. A benchmark:

$$\sigma(s = 9m^2) = \frac{g^4}{288\pi m^6}\left(\frac{1}{64} + \frac{1}{3} + \frac{\ln 6}{70}\right) = 4.1397 \times 10^{-4}\,\frac{g^4}{m^6}. \tag{10.13.6}$$

Limits

From (10.13.5),

$$\sigma \to \frac{25\,g^4}{1152\pi m^6} = \frac{|\mathcal{T}_{\rm thr}|^2}{32\pi s} \quad (s \to 4m^2), \qquad \sigma \to \frac{g^4}{16\pi m^2 s^2} \quad (s \gg m^2). \tag{10.13.7}$$

At threshold σ stays finite although the final momenta vanish: $|\mathbf{k}_1'|$ in $d\mathrm{LIPS}_2$ and $|\mathbf{k}_1|$ in the flux factor go to zero together, and the isotropic $\mathcal{T}_{\rm thr}$ gives $\sigma = \frac{1}{2}\cdot 4\pi\, d\sigma/d\Omega$.
At high energy (the exact σ is 1.037 and 1.0004 times the limit at $s = 10^2 m^2$ and $10^4 m^2$) the s-channel term of $\mathcal{T}$ is $O(g^2/s)$ and drops out. The t- and u-channel poles, at $t = m^2$ and $u = m^2$, lie just beyond the forward edge $t = 0$ and the backward edge $u = 0$ of the physical region. Near $t = 0$, $d\sigma/dt \approx g^4/[16\pi s^2(m^2 - t)^2]$, and $\int_0^\infty dx/(m^2 + x)^2 = 1/m^2$ gives the limit after the $\frac{1}{2}$ and the equal backward peak. So σ comes from a forward and a backward peak, about $2m/\sqrt{s}$ wide in angle (where $|t| = m^2$): at $s = 10^4 m^2$, 91% of it has $|t| < 10m^2$ or $|u| < 10m^2$.
As $m \to 0$ at fixed s, $\sigma \approx g^4/(16\pi m^2 s^2) \to \infty$. At $m = 0$ the two poles sit on the edges of the physical region, $d\sigma/dt \propto 1/t^2$ near $t = 0$, and the integral diverges at forward and backward angles, while $d\sigma/d\Omega$ stays finite at every $\theta \neq 0, \pi$. What massless particles do to observable cross sections is taken up later (LecNote15, §26).

Units: from g^4/m^6 to barns

σ comes out in mass^{-2}; multiply by $(\hbar c)^2$ to get an area. The SI values of h, c and e are exact, and $\hbar c = hc/2\pi$, converted from J m to eV m by dividing by e, is $197.3269804\ldots$ MeV fm. With $1\ \mathrm{b} = 10^{-28}\ \mathrm{m}^2 = 100\ \mathrm{fm}^2$, so $1\ \mathrm{fm}^2 = 10$ mb,

$$(\hbar c)^2 = 0.389379\ \mathrm{GeV}^2\,\mathrm{mb}, \qquad 1\ \mathrm{GeV}^{-2} = 0.3894\ \mathrm{mb}. \tag{10.13.8}$$

For $g = m = 1$ GeV and $\sqrt{s} = 3$ GeV, (10.13.6) gives $\sigma = 4.1397 \times 10^{-4}\ \mathrm{GeV}^{-2} = 161.2$ nb. In general $\sigma(9m^2) = 161.2\ \mathrm{nb} \times (g/m)^4 (1\ \mathrm{GeV}/m)^2$.

A two-body decay

In φ^3 theory alone nothing decays: two or more φ's weigh at least $2m > m$. Add a scalar Φ of mass M with $\mathcal{L}_1 \supset \frac{h}{2}\Phi\varphi^2$, h a mass. The two φ lines attach to φ^2 in $2!$ ways, so the vertex is ih and at tree level $\mathcal{T}(\Phi \to \varphi\varphi) = h$. For $M > 2m$, in the Φ rest frame ($\sqrt{s} \to M$, $E_1 \to M$), $|\mathbf{k}'| = \frac{M}{2}\sqrt{1 - 4m^2/M^2}$ and $\int_{4\pi} d\mathrm{LIPS}_2 = \sqrt{1 - 4m^2/M^2}/(8\pi)$. The two φ's are identical, so the $\frac{1}{2}$ applies again:

$$\Gamma = \frac{1}{2M}\cdot\frac{1}{2}\cdot h^2\,\frac{\sqrt{1 - 4m^2/M^2}}{8\pi} = \frac{h^2}{32\pi M}\sqrt{1 - \frac{4m^2}{M^2}}. \tag{10.13.9}$$

The $2!$ and the $\frac{1}{2}$ are different things and both stay: one counts contractions in a diagram, the other final states that the phase-space integral counts twice.

In a frame where Φ has energy E, $\int |\mathcal{T}|^2 d\mathrm{LIPS}$ is the same scalar, a function of $k_1^2 = -M^2$ only, and only the $1/2E_1$ changes:

$$\Gamma_{\text{lab}} = \Gamma \frac{M}{E} = \frac{\Gamma}{\gamma}, \qquad \tau_{\text{lab}} = \gamma\tau, \tag{10.13.10}$$

which is time dilation; the invariant is the proper width, p. 10's red formula. Doing the phase-space integral directly in the moving frame, with no rest-frame input, reproduces it, also when Φ moves faster than its decay products do in its rest frame, so that both go into a forward cone, with two values of $|\mathbf{k}_1'|$ in each direction.

The same vertex puts Φ in the s-channel of $\varphi\varphi \to \varphi\varphi$, with denominator $M^2 - s - i\epsilon$. For $M > 2m$, exactly when Φ can decay, $s = M^2$ is in the physical region: the $i\epsilon$ can no longer be dropped, and the tree-level σ diverges there. Treating it needs the width of Φ: a resonance, Srednicki's §25, not covered in these notes.

Summary

- In the physical region the s-channel term of (10.13.2) is negative, the others positive; no pole is reached, so the $i\epsilon$ drops. The $\frac{1}{2}$ for identical particles goes into σ only; $d\sigma/d\Omega = |\mathcal{T}|^2/(64\pi^2 s)$ is what a detector counts.
- σ is (10.13.5): $4.1397 \times 10^{-4} g^4/m^6$ at $s = 9m^2$, finite at threshold, $g^4/(16\pi m^2 s^2)$ at high energy from the forward and backward peaks, infinite at $m = 0$. $1\ \mathrm{GeV}^{-2} = 0.3894$ mb.
- $\Gamma(\Phi \to \varphi\varphi) = h^2\sqrt{1 - 4m^2/M^2}/(32\pi M)$, with the vertex's $2!$ and the phase space's $\frac{1}{2}$ both kept; $\Gamma_{\text{lab}} = \Gamma M/E$.